

2920/205
VISUAL PROGRAMMING
November 2017
Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE II
VISUAL PROGRAMMING

3 hours

INSTRUCTIONS TO CANDIDATES

*Answer any **FIVE** of the following **EIGHT** questions in the answer booklet provided.*
All questions carry equal marks
Candidates should answer the questions in English.

This paper consists of 4 printed pages.
Candidates should check the question paper to ascertain that all the
pages are printed as indicated and that no questions are missing.

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Turn over

1.
 - (a)
 - (i) Outline **three** factors that could be considered when setting control properties in a Visual Basic program. (3 marks)
 - (ii) Distinguish between an *object* and a *control* as used in Visual Basic program Environment (2 marks)
 - (b)
 - (i) Describe the term *menu editor* as used in Visual Basic programming. (2 marks)
 - (ii) Describe **two** properties applied in a menu editor. (4 marks)
 - (c)
 - (i) Distinguish between *sentinel* and *flag controlled loops* as used in programming. (4 marks)
 - (ii) Write a program in Visual Basic that would display even numbers between 10 and 20 on a form. Use a *do while* statement. Attach the code to a command button. (5 marks)
2.
 - (a) State the *prefix* for each of the following Visual Basic tools: (2 marks)
 - (i) Timer;
 - (ii) Option Button;
 - (iii) Shape;
 - (iv) Horizontal ScrollBar.
 - (b) Describe **three** Project Explorer buttons used in a Visual Basic program. (6 marks)
 - (c) Figure 1 shows a list of numbers in an array. Use it to answer the question that follows.

11	12	17	19	21	25	26	28
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Figure 1

Perform a binary search on the array showing the steps, given that the search key is 28. (5 marks)
 - (d) Write a program in Visual Basic that accepts a name of a person through the use of an input box. The program should then add the name to a list box through the use of a procedure. Attach the code to a command button. (7 marks)
3.
 - (a) Outline **four** characteristics of a Visual Programming language. (4marks)
 - (b)
 - (i) Describe a sub procedure as used in a Visual programming. (2 marks)
 - (ii) Outline **three** characteristics of an event as used in Visual Basic programming. (3 marks)
 - (c) Distinguish between *unload* and *hide methods* as used in a Visual Basic program. (4 marks)

- (d) A book supplier offers a discount of 10% when the total cost of books purchased is more than 10,000. Write a program in Visual Basic that accepts the price of three books one at a time through an input box. The program computes total cost, discount if any and then displays the amount payable on a picture box. Attach the code to a command button.

(7 marks)

4. (a) The following are programming languages.

Visual Basic, PHP JavaScript, Visual C++, Python, lisp

Categorise each language as either event driven or procedural programming languages.

(3 marks)

- (b) Outline **three** properties of a user defined function. (3 marks)
- (c) Explain **three** similarities between *ADO data control* and *data environment* as used in Visual Basic programming. (6 marks)
- (d) (i) Explain the use of *ubound()* as used in an array. (2 marks)
- (ii) Write a program in Visual Basic that accepts five values in an array. The program then displays the smallest and the largest value on the form. Attach the code to a command button. (6 marks)

- 5 (a) State the output generated when each of the following Visual Basic statement is executed. (4 marks)

- (i) Mid ("Visual Environment", 3, 6)
- (ii) InStr (1, "Visual Environment", "Environment")
- (iii) Print Len ("Visual Environment")
- (iv) Print RTrim("Visual Environment ")

- (b) Describe **two** values for an EOF action in a Visual Basic program. (4 marks)

- (c) John, a programmer would like to include *error handling messages* in a Visual Basic program. Explain **three** approaches that he could consider. (6 marks)

- (d) Table 1 shows details of Module 1 subjects. Use it to answer the question that follows:

Subject code	Subject name	Label colour
4	ICT	Red
6	Programming	Green
8	Mathematics	Orange
Any other number	Does not exist	White

Write a program in Visual Basic that accepts the subject code through the use of a textbox. The program then displays the subject name on a label and changes the label background colour as indicated when a command button is clicked.

Use a *select case* statement.

(6 marks)

- 6 (a) Outline **three** *Recordset* methods that enables navigation in a Visual Basic program. (3 marks)
- (b) Describe **three** *ADO data locking* options used in a Visual Basic program. (6 marks)
- (c) Alice tried connecting a database table to a form. When she clicked the test connection button, a message "unsuccessful" was displayed. Explain **three** possible reasons that could lead to this. (6 marks)
- (d) Write a program in Visual Basic that would accept the name of a person through the use of a text box. The program then displays "Hallo Nelly" on a message box() when the name entered is Nelly, otherwise it displays "unknown name". Attach the code to a command button. (5 marks)
- 7 (a) Outline **two** *keypress* events and **two** *mouse events* in a Visual Basic program. (4 marks)
- (b) Differentiate between *Print x, y* and *Picture1.Print x, y* statements as used in a Visual Basic program. (4 marks)
- (c) Marion, a programmer would like to include *err object* method in a Visual Basic program. Explain **two** methods that she could consider. (4 marks)
- (d) Write a program in Visual Basic that uses a procedure to capture the radius of a circle through the use of a textbox. The program then computes the area of a circle and volume of a sphere through a procedure and displays the results on a form. Attach the code to a command button. (8 marks)
- 8 (a) Outline **four** sorting techniques that could be used in a Visual Basic program. (4 marks)
- (b) Mary prefers using a binary tree to search for a record. Explain **three** challenges she could face. (6 marks)
- (c) Figure 2 shows a user interface in a Visual Basic program. Use it to answer the question that follows.

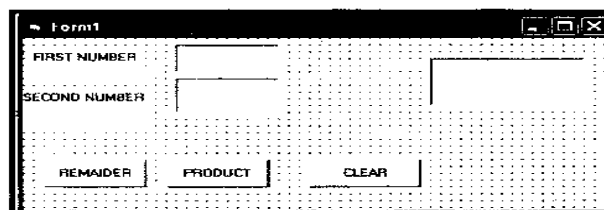


Figure 2

Write a program in Visual Basic that could implement the interface. The program should clear the results in the *picturebox* when the button clear is clicked. (10 marks)

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